

MODULE 12 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

Compute

From a bare instance to a function. Choosing the right point on the spectrum.

DOMAIN 3 of 4 EXAM WEIGHT 34% MODULE 12 of 26

Every compute question is asking the same thing: **how much of the stack do you want to manage?**

The compute spectrum

SERVICE	YOU MANAGE	AWS MANAGES
Amazon EC2	OS, patching, scaling, capacity	Hardware and hypervisor
ECS or EKS on EC2	The container hosts	Orchestration control plane
AWS Fargate	Container images only	All the underlying compute
Elastic Beanstalk	Application code	Environment provisioning
AWS Lambda	Function code only	Everything else

ON THE EXAM

Read the middle column. As it shortens, operational overhead falls and control falls with it.

Amazon EC2

- ◆ **Virtual servers in the cloud.** You choose the operating system, size and network placement.
- ◆ You are responsible for the guest OS, patching, and everything installed on it.
- ◆ **Billed per second** for most Linux instances, with a sixty-second minimum.
- ◆ Stopping an instance stops compute charges. Attached EBS storage still costs money.
- ◆ Terminating destroys the instance. Stopping preserves it. The exam distinguishes these carefully.

ON THE EXAM

Stopped means no compute charge but storage still bills. Terminated means gone. That distinction appears often.

Instance families and AMIs

- ◆ **General purpose** balances compute, memory and networking. The default starting point.
- ◆ **Compute optimised** for processor-heavy work. **Memory optimised** for large in-memory datasets.
- ◆ **Storage optimised** for high local disk throughput. **Accelerated computing** for GPUs.
- ◆ You are not expected to know instance type names or specifications. Recognising the categories is enough.
- ◆ **An Amazon Machine Image** is the template an instance launches from. It is Regional.

ON THE EXAM

Match the bottleneck to the family. CPU-bound means compute optimised; a large dataset in memory means memory optimised.

EC2 or Lambda?

AMAZON EC2

A server that keeps running

- Runs continuously until you stop it
- Any runtime, any software, long-running work
- You patch and scale it
- Billed for the time it runs, used or not

AWS LAMBDA

Code that runs on an event

- Runs when something triggers it, then stops
- Maximum fifteen minutes per invocation
- No servers to patch or scale
- Billed per request and duration only

THE TIEBREAKER

Continuous or longer than fifteen minutes means EC2. Event-driven and short means Lambda.

AWS Lambda in detail

- ◆ **Fifteen minutes is the hard maximum** for a single invocation. Longer work needs something else.
- ◆ **It scales automatically** from zero to many concurrent executions with no configuration.
- ◆ You pay nothing when it is not running, which suits spiky or infrequent workloads.
- ◆ Triggered by S3 uploads, API Gateway requests, EventBridge schedules, queue messages and many others.
- ◆ It is stateless. Anything that must persist goes to S3, DynamoDB or another store.

ON THE EXAM

The fifteen-minute limit is the most frequently decisive fact about Lambda on this exam.

Containers on AWS

- ◆ **Amazon ECS** is AWS's own container orchestrator. Simpler, deeply integrated with AWS.
- ◆ **Amazon EKS** is managed Kubernetes. Choose it when Kubernetes specifically is required.
- ◆ **Amazon ECR** is the registry where container images are stored. It orchestrates nothing.
- ◆ Both ECS and EKS can run on EC2 instances you manage, or on Fargate.
- ◆ ECR is a common distractor for questions about running containers. It only stores images.

ON THE EXAM

Deploy and scale containers means ECS or EKS. Store container images means ECR. They are not alternatives.

ECS or EKS?

AMAZON ECS

AWS's own orchestrator

- Simpler to learn and operate
- Tight integration with AWS services
- No Kubernetes knowledge needed
- The default unless Kubernetes is named

AMAZON EKS

Managed Kubernetes

- Standard, portable Kubernetes API
- Existing Kubernetes tooling and skills transfer
- More moving parts to understand
- Choose when the question names Kubernetes

THE TIEBREAKER

If the scenario says Kubernetes, or mentions existing Kubernetes skills or workloads, the answer is EKS.

Fargate or the EC2 launch type?

AWS FARGATE

Serverless containers

- No instances to provision or patch
- Pay for the resources each task requests
- AWS handles capacity entirely
- Less control over the underlying host

EC2 LAUNCH TYPE

Your own container hosts

- You provision and patch the instances
- Pay for the instances, used or idle
- Control over instance type and placement
- Cheaper at high, steady utilisation

THE TIEBREAKER

No servers to manage means Fargate. Needing control of the host, or already-paid-for capacity, means EC2.

AWS Batch and Amazon Lightsail

- ◆ **AWS Batch runs large numbers of batch computing jobs.** It provisions the compute automatically and queues the work.
- ◆ The signal is thousands of jobs, overnight processing, or scientific and financial computation.
- ◆ **Amazon Lightsail is a simplified virtual private server** with a predictable monthly price.
- ◆ It bundles compute, storage and networking, and offers pre-configured blueprints such as WordPress.
- ◆ The signal is a simple application, a small team, and a wish to avoid AWS complexity.

ON THE EXAM

Thousands of scheduled jobs means Batch. Simple, cheap and predictable for a small workload means Lightsail.

Amazon EC2 Auto Scaling

- ◆ **It adds and removes instances** to match demand, within a minimum and maximum you set.
- ◆ It also replaces unhealthy instances automatically, which is a reliability benefit as much as a cost one.
- ◆ **Dynamic scaling** responds to a metric. **Scheduled scaling** follows a known pattern. **Predictive scaling** forecasts.
- ◆ It works with Elastic Load Balancing so new instances receive traffic automatically.
- ◆ Auto Scaling changes the number of instances. It never changes the size of one.

ON THE EXAM

Count, not size. Resizing an instance is rightsizing, and Compute Optimizer recommends it.

Elastic Load Balancing

- ◆ **Application Load Balancer** works at layer 7 with HTTP and HTTPS, routing on path or hostname.
- ◆ **Network Load Balancer** works at layer 4 with TCP and UDP, for extreme performance and static IP addresses.
- ◆ **Gateway Load Balancer** sits in front of third-party virtual appliances such as firewalls.
- ◆ **Classic Load Balancer** is the previous generation. Still supported in a VPC, but never the right answer for a new design.
- ◆ All types perform health checks and only send traffic to healthy targets.

ON THE EXAM

Older question banks still offer Classic Load Balancer. It works, but AWS recommends ALB or NLB for anything new.

Application or Network Load Balancer?

APPLICATION LOAD BALANCER

Understands the request

- HTTP and HTTPS traffic
- Routes on URL path, hostname or header
- Integrates with AWS WAF
- The default for web applications

NETWORK LOAD BALANCER

Moves packets fast

- TCP, UDP and TLS traffic
- Extreme throughput and very low latency
- Provides a static IP address per AZ
- For non-HTTP protocols and gaming

THE TIEBREAKER

Routing based on the content of a request means ALB. Static IP, non-HTTP protocols, or extreme performance means NLB.

Reverse lookup

IF THE SCENARIO SAYS	THE SERVICE IS
Run code on an event, no servers, short duration	AWS Lambda
Full control of the operating system	Amazon EC2
Run containers without managing servers	AWS Fargate
Kubernetes, or existing Kubernetes skills	Amazon EKS
Thousands of jobs processed overnight	AWS Batch
Simple workload, predictable monthly price	Amazon Lightsail
Handle traffic increases automatically	EC2 Auto Scaling
Deploy an app without managing infrastructure	AWS Elastic Beanstalk

ON THE EXAM

Eight phrasings covering most compute questions on the exam. These wordings barely vary.

A company needs to generate a thumbnail whenever an image is uploaded to an Amazon S3 bucket. Uploads arrive unpredictably, sometimes hundreds in a minute and sometimes none for hours. Each thumbnail takes about four seconds to produce.

The company does not want to manage any servers.

Which service should the company use?

- | | |
|---|---|
| ☐ A Amazon EC2 with Auto Scaling | ☐ B AWS Lambda |
| ☐ C AWS Batch | ☐ D Amazon Lightsail |

◆ PAUSE AND DECIDE

Which service should the company use?

- A

 Amazon EC2 with Auto Scaling
Works, but instances run and bill during the quiet hours, and you still patch them.
- B

 AWS Lambda
Triggered directly by the S3 upload, scales from zero, four seconds is well inside the limit, nothing to manage.
- C

 AWS Batch
For large volumes of scheduled batch jobs, not for reacting to a single upload immediately.
- D

 Amazon Lightsail
A simplified server you still manage. It does not respond to events.

THE SIGNAL

Three signals: triggered by an upload, unpredictable volume, and no servers. All three point at Lambda.

A research group runs a simulation that takes roughly forty minutes per run. Each night it needs to complete about two thousand of these runs, and the results are not needed until the following morning.

The runs are independent and can be retried if one fails.

Which service is most appropriate?

- | | |
|--|---|
| ☐ A AWS Lambda | ☐ B AWS Batch |
| ☐ C AWS Elastic Beanstalk | ☐ D Amazon Lightsail |

◆ PAUSE AND DECIDE

Which service is most appropriate?

A AWS Lambda

Forty minutes exceeds the fifteen-minute maximum. Ruled out by a hard limit, not a preference.

C AWS Elastic Beanstalk

Runs web application environments. It is not a job scheduler.

B AWS Batch

Purpose-built for large numbers of independent jobs. It queues the work and provisions compute automatically.

D Amazon Lightsail

A single simplified server. Two thousand forty-minute runs a night is far beyond its purpose.

THE SIGNAL

Forty minutes eliminates Lambda outright. Thousands of independent jobs is the Batch signature.

A company runs containerised microservices. Its platform team has years of Kubernetes experience, maintains a large library of Kubernetes manifests, and wants to keep using its existing tooling.

It does not want to operate the Kubernetes control plane itself.

Which service meets these requirements?

- | | |
|---------------------------------------|---|
| ☐ A Amazon ECS | ☐ B Amazon EKS |
| ☐ C Amazon ECR | ☐ D Kubernetes on self-managed EC2 instances |

◆ PAUSE AND DECIDE

Which service meets these requirements?

- A

 Amazon ECS
A capable orchestrator, but it does not run Kubernetes manifests. The existing library and tooling would be wasted.
- B

 Amazon EKS
Managed Kubernetes. Existing manifests and tooling work, and AWS operates the control plane.
- C

 Amazon ECR
Stores container images. It runs nothing.
- D

 Kubernetes on self-managed EC2 instances
Gives them Kubernetes, but they explicitly do not want to operate the control plane.

THE SIGNAL

Existing Kubernetes manifests and tooling rules out ECS. Not operating the control plane rules out self-managed.

A team already runs containers on Amazon ECS using EC2 instances it manages. Patching those instances and forecasting how many are needed has become the team's largest operational burden.

The workload itself is not changing, and the team wants to keep using ECS.

What should the team do?

- ☐ A Migrate from Amazon ECS to Amazon EKS
- ☐ B Switch the ECS tasks to the AWS Fargate launch type
- ☐ C Rewrite the containers as AWS Lambda functions
- ☐ D Add an Auto Scaling group for the EC2 instances

◆ PAUSE AND DECIDE

What should the team do?

- A

Migrate from Amazon ECS to Amazon EKS
Changes orchestrator without removing the hosts. More work, same burden, and the team wants to keep ECS.
- B

Switch the ECS tasks to the AWS Fargate launch type
Same ECS, same containers, but no instances to patch or size. The burden disappears.
- C

Rewrite the containers as AWS Lambda functions
A rewrite of a workload the scenario says is not changing. Disproportionate.
- D

Add an Auto Scaling group for the EC2 instances
Helps with forecasting capacity, but the instances still exist and still need patching.

THE SIGNAL

Two burdens named: patching and capacity forecasting. Only Fargate removes both, and keeps ECS.

A charity wants to host a WordPress site. It has one part-time technical volunteer, needs a predictable monthly bill for its budget, and does not want to configure networking, load balancing or scaling.

Traffic is modest and steady.

Which service best fits?

- | | |
|--|---|
| ☐ A Amazon EC2 with an Auto Scaling group | ☐ B Amazon Lightsail |
| ☐ C AWS Elastic Beanstalk | ☐ D Amazon EKS |

◆ PAUSE AND DECIDE

Which service best fits?

- A

 Amazon EC2 with an Auto Scaling group
Requires exactly the networking and scaling configuration the charity said it does not want.
- B

 Amazon Lightsail
Bundled compute, storage and networking at a fixed monthly price, with a WordPress blueprint ready to launch.
- C

 AWS Elastic Beanstalk
Closer, but it provisions EC2 and load balancers whose costs vary, and it targets code you wrote.
- D

 Amazon EKS
A Kubernetes cluster for a WordPress site with one volunteer. Wildly disproportionate.

THE SIGNAL

Predictable monthly bill is the Lightsail giveaway. Bundled pricing is its defining feature.

An online retailer's web tier runs on EC2 instances behind a load balancer. Traffic triples every weekday lunchtime and falls away overnight. The company currently runs enough instances for the lunchtime peak all day.

It wants to reduce cost without risking availability at peak.

What should the retailer do?

- ☐ A Replace the instances with larger ones
- ☐ B Configure EC2 Auto Scaling to add and remove instances with demand
- ☐ C Move the web tier to AWS Lambda
- ☐ D Purchase Reserved Instances for the peak capacity

◆ PAUSE AND DECIDE

What should the retailer do?

- A

 Replace the instances with larger ones
Increases cost and still runs peak capacity all day. It changes size when the problem is count over time.
- B

 Configure EC2 Auto Scaling to add and remove instances with demand
Capacity follows the traffic curve. Peak is still covered, and quiet hours cost less.
- C

 Move the web tier to AWS Lambda
Would scale, but re-architecting a working web tier is far beyond what the scenario asks.
- D

 Purchase Reserved Instances for the peak capacity
Discounts the over-provisioning rather than removing it. Cheaper waste is still waste.

THE SIGNAL

A predictable daily curve with over-provisioning all day is the textbook Auto Scaling scenario.

A company runs a multiplayer game server that communicates over UDP. It needs to distribute traffic across many instances, requires extremely low latency, and must present a fixed IP address that players' clients are configured to reach.

It also runs a separate web portal that routes requests to different services based on the URL path.

Which load balancers should the company use? (Choose two.)

- ☐ A Network Load Balancer for the game server
- ☐ B Application Load Balancer for the game server
- ☐ C Application Load Balancer for the web portal
- ☐ D Classic Load Balancer for the web portal
- ☐ E Gateway Load Balancer for both

◆ PAUSE AND DECIDE

Which load balancers should the company use? (Choose two.)

- A

 Network Load Balancer for the game server
UDP, extreme low latency and a static IP address. Three NLB signatures in one requirement.
- B

 Application Load Balancer for the game server
ALB handles HTTP and HTTPS only. It cannot carry UDP at all.
- C

 Application Load Balancer for the web portal
Routing on URL path is layer 7 content-based routing, which is what ALB is for.
- D

 Classic Load Balancer for the web portal
Previous generation. It works, but AWS recommends ALB and the exam expects the current answer.
- E

 Gateway Load Balancer for both
For deploying third-party virtual appliances such as firewalls. Neither workload is an appliance.

THE SIGNAL

UDP and a static IP force NLB. Routing on URL path forces ALB. Each requirement names its own layer.

A financial firm must run a legacy application that requires a specific, unsupported operating system version, needs full administrative access to configure kernel parameters, and must run continuously.

The firm asks which AWS compute option is suitable.

Which service should the firm use?

- | | |
|---------------------------------------|--|
| ☐ A AWS Lambda | ☐ B AWS Fargate |
| ☐ C Amazon EC2 | ☐ D AWS Elastic Beanstalk |

◆ PAUSE AND DECIDE

Which service should the firm use?

- A

 AWS Lambda
No operating system access, supported runtimes only, and a fifteen-minute limit. Fails on all three requirements.
- B

 AWS Fargate
No access to the host or its kernel. Containers share the host kernel, which cannot be configured per task.
- C

 Amazon EC2
Full control of the operating system, any OS image, and it runs continuously. The only option that qualifies.
- D

 AWS Elastic Beanstalk
Supports specific managed platforms. An unsupported OS version is outside what it provisions.

THE SIGNAL

Kernel-level configuration is the strongest possible signal for EC2. Managed services take that access away.

What to carry out of this module

- ◆ **Ask how much of the stack the company wants to manage.** The scenario always says.
- ◆ **Fifteen minutes is Lambda's hard limit.** Check any stated duration against it first.
- ◆ Kubernetes named means EKS. Otherwise ECS. ECR only stores images.
- ◆ Fargate removes the hosts. Auto Scaling changes the count, never the size.
- ◆ **URL path routing means ALB. UDP, static IP or extreme performance means NLB.**

ON THE EXAM

Purchase options — On-Demand, Reserved, Spot and Savings Plans — are taught in Module 18, not here.